TRAINING MANAGEMENT SYSTEM (TMS) – FINAL DOCUMENTATION

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1. Project Overview

Project Title:
Training Management System (TMS)

Description:
The Training Management System (TMS) is a web-based, role-driven application designed to manage short-term training programs conducted in **weekly batches**. The system centralizes operations such as student management, trainer allocation, attendance tracking, performance evaluation, exam scheduling, and certification.

The system follows a **Batch-Centric Architecture**, where each batch acts as an independent unit containing all operational data.

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2. Objectives

- Automate training program management
- Implement batch-wise structured workflow
- Enable centralized control with role-based access
- Track performance of students and trainers
- Provide analytical insights for decision-making

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3. Core Concept: Batch-Centric System

Batch as the Core Entity

Each batch contains:

- Students
- Trainers
- Labs
- Attendance
- Exams
- Results
- Certifications

Example:

- Batch 1 → 1 Week
- Batch 2 → Next Week

All operations are **isolated within a batch**

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4. User Roles & Responsibilities

Admin

- Add/remove Managers
- Add/remove College Staff
- Full system access
- Monitor system performance

Manager (Main Controller)

- Create batches (start date & end date)
- Add/update/remove students
- Add/manage trainers
- Assign:
  - Trainers → Labs
  - Students → Labs & Batches
- Mark trainer attendance
- Mark holidays
- Schedule exams (date, slots, labs)
- Assign certificate voucher codes
- Control College Staff permissions

Trainer

- Add student details
- Mark student attendance
- Add:
  - Mid Test Results
  - Final Mock Results
- Set passing criteria
- Evaluate students (auto pass/fail)
- Assign course & certification

College Staff

- View-only access
- Permissions controlled by Manager
- Can view:
  - Students
  - Attendance
  - Results
  - Reports

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5. Access Control System

Role-Based + Permission-Based Access

- Admin, Manager, Trainer → Role-Based
- College Staff → Permission-Based

Example Permissions:

- VIEW_STUDENTS
- VIEW_ATTENDANCE
- VIEW_RESULTS
- VIEW_REPORTS

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6. System Modules

6.1 Authentication Module

- Secure login system
- JWT-based authentication
- Role-based authorization

6.2 Batch Management Module

- Create batch
- Define:
  - Start Date
  - End Date
- Track batch status:
  - Active / Completed

6.3 Student Management Module

- Add/update/remove students
- Store personal details
- Assign students to:
  - Batch
  - Lab

6.4 Trainer Management Module

- Add trainers
- Assign trainers to labs
- Track trainer attendance

6.5 Lab Management Module

- Create labs within a batch
- Assign:
    - Trainer
    - Students

6.6 Attendance Management

Student Attendance:

- Mark daily attendance (batch-wise)

Trainer Attendance:

- Present
- Late
- Half Day
- Absent
- Leave

Holiday Management:

- Manager can declare holidays

6.7 Performance & Evaluation Module

- Mid Test results
- Final Mock results
- Passing criteria (%)
- Automatic pass/fail calculation
- Student filtering

6.8 Exam Scheduling Module

- Set exam date
- Define time slots
- Assign labs
- Allocate students to slots

6.9 Certification Module

- Assign certification name
- Generate/assign voucher codes
- Track pass/fail status

6.10 Analytics & Dashboard Module

Features:

- Trainer performance analysis
- Pass vs Fail ratio
- Batch-wise performance
- Ranking system
- Trend analysis

6.11 Reporting Module

- Generate reports:
  - Student performance
  - Trainer performance
  - Attendance
- Export to PDF/Excel

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7. Database Design (High-Level)

Core Collections:

- Users
- Batches
- Students
- Trainers
- Labs
- Attendance
- Exams
- Results
- Certificates
- Permissions

Important Rule:

All major entities include:
batchId

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8. System Workflow

Phase 1: Setup

- Admin creates Manager & Staff

Phase 2: Batch Creation

- Manager creates batch with:
  - Start date
  - End date

Phase 3: Resource Allocation

- Manager assigns:
  - Students → Batch & Labs
  - Trainers → Labs

Phase 4: Daily Operations

- Trainer marks attendance
- Manager marks trainer attendance & holidays

Phase 5: Evaluation

- Trainer adds test results
- System calculates pass/fail

Phase 6: Exam Scheduling

- Manager schedules exam (date, slots, labs)

Phase 7: Final Results

- Trainer/Manager enters marks
- System calculates final result

Phase 8: Certification

- Manager assigns voucher codes

Phase 9: Analytics

- System generates dashboards

Phase 10: Batch Completion

- Batch marked as completed
- New batch created

□ 9. Dashboard Features

- Trainer ranking
- Pass vs Fail statistics
- Batch performance
- Student performance trends
- Attendance insights

□ 10. Additional Features

- Notification system
- Audit logs
- Search & filtering
- Responsive UI
- Dark/light mode

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11. Technology Stack

Frontend:

- React.js
- Tailwind CSS

Backend:

- Node.js
- Express.js

Database:

- MongoDB

Authentication:

- JWT

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12. Security Features

- Password hashing (bcrypt)
- JWT authentication
- Role & permission validation
- Secure APIs

WORKFLOW

Phase 1: Batch Creation (Manager Starts Everything)

Manager First Action:

1. Create a new batch:
    - Batch Name (Batch 1, Batch 2…)
    - Start Date
    - End Date

Example:

- Batch 1 → 1st Jan – 7th Jan
- Batch 2 → 8th Jan – 14th Jan

Phase 2: Everything Happens Inside the Batch

Once batch is created

Student Allocation (Inside Batch)

- Add students to **specific batch**
- Assign students to:
    - Labs (within that batch)

Trainer Allocation (Inside Batch)

- Assign trainers to:
  - Labs (within that batch)

Lab Setup (Inside Batch)

- Create labs under batch
- Each lab has:
  - Trainer
  - Students

Phase 3: Daily Operations (Inside Batch)

Trainer:

- Marks **student attendance (batch-specific)**
- Adds **test results (batch-specific)**

Manager:

- Marks **trainer attendance (batch-specific)**
- Declares **holidays (batch-specific)**

Phase 4: Evaluation (Inside Batch)

Trainer:

- Adds:
  - Mid Test
  - Final Mock

System:

- Calculates:
  - Pass/Fail
- Filters students

Phase 5: Exam Scheduling (Inside Batch)

Manager:

- Sets:
  - Exam Date (within batch duration)
  - Time Slots
  - Labs
- Assigns students to slots

Phase 6: Final Exam & Results (Inside Batch)

Trainer / Manager:

- Enter final marks

System:

- Calculates final:
  - Pass/Fail

Phase 7: Certification (Inside Batch)

Manager:

- Assigns:
  - Voucher codes
  - Certification

Phase 8: Analytics (Batch-Based)

Dashboard shows:

- Batch Performance
- Trainer Performance (within batch)
- Pass/Fail ratio
- Top students

Phase 9: Batch Lifecycle Ends

When:

- End date is reached

Batch status:

- Active → Completed

Data remains stored for reports

Phase 10: New Batch Starts

- Manager creates next batch
- Same workflow repeats